THE INDUCED TOPOLOGY ON A SUBCATEGORY FROM THE BIG PH TOPOLOGY

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2796, 2810] and proof(s) [[2812, 2838]]; this spec covers the complete environment instead (statement 2796-2810, proof span(s) [(2812, 2838)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original definition 0DBJ: sources/stacks-chow/tags.txt, source lines 16699–16699. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DBH: sources/stacks-chow/tags.txt, source lines 16697–16697. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DBI: sources/stacks-chow/tags.txt, source lines 16698–16698. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 000R: sources/stacks-chow/tags.txt, source lines 36–36. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 000X: sources/stacks-chow/tags.txt, source lines 42–42. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let Sch_{ph} be a big ph site as in Definition 0DBJ. Let $T \in \mathrm{Ob}(Sch_{ph})$. Let $\{T_i \to T\}_{i\in I}$ be an arbitrary ph covering of T.*

1. *There exists a covering $\{U_j \to T\}_{j\in J}$ of T in the site Sch_{ph} which refines $\{T_i \to T\}_{i\in I}$.*
2. *If $\{T_i \to T\}_{i\in I}$ is a standard ph covering, then it is tautologically equivalent to a covering of Sch_{ph}.*
3. *If $\{T_i \to T\}_{i\in I}$ is a Zariski covering, then it is tautologically equivalent to a covering of Sch_{ph}.*

Proof. For each i choose an affine open covering $T_i = \bigcup_{j\in J_i} T_{ij}$ such that each T_{ij} maps into an affine open subscheme of T. By Lemmas 0DBH and 0DBI the refinement $\{T_{ij} \to T\}_{i\in I, j\in J_i}$ is a ph covering of T as well. Hence we may assume each T_i is affine, and maps into an affine open W_i of T. Applying Sets, Lemma 000R we see that W_i is isomorphic to an object of Sch_{ph}. But then T_i as a finite type scheme over W_i is isomorphic to an object V_i of Sch_{ph} by a second application of Sets, Lemma 000R. The covering $\{V_i \to T\}_{i\in I}$ refines $\{T_i \to T\}_{i\in I}$ (because

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they are isomorphic). Moreover, $\{V_i \to T\}_{i \in I}$ is combinatorially equivalent to a covering $\{U_j \to T\}_{j \in J}$ of T in the site Sch_{ph} by Sets, Lemma 000R. The covering $\{U_j \to T\}_{j \in J}$ is a refinement as in (1). In the situation of (2), (3) each of the schemes T_i is isomorphic to an object of Sch_{ph} by Sets, Lemma 000R, and another application of Sets, Lemma 000X gives what we want. $\square$